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# Corporate Integrity Standard - Piping Inspection and Integrity

## Scope and Application

This CorporateStandard establishes the mandatory framework for piping inspection and integrity management across all QSCP facilities. It defines the requirements for digital data management, Condition Monitoring Location (CML) identification, and advanced non-destructive testing methodologies. This document replaces QSCP-CIS-001 Rev 0, which previously established the baseline for Risk-Based Inspection (RBI) intervals and Fitness-For-Service (FFS) assessments.

Compliance with this standard ensures that all piping systems are managed in accordance with API 570. The requirements specified herein apply to all metallic and non-metallic piping circuits, focusing on the standardization of CML-id rules and the integration of digital data into the corporate asset integrity management system.

## Personnel Qualifications and Responsibilities

Dr. Helen Marchetti, as the Corporate Asset Integrity Manager, is responsible for the technical governance and oversight of this standard. All personnel performing non-destructive examinations must hold valid certifications. Specifically, technicians must be certified to CAN/CGSB-48.9712 or EN ISO 9712 to ensure competency in the execution of inspection tasks and data interpretation.

The Corporate Asset Integrity department is tasked with auditing site compliance and ensuring that qualified engineers perform FFS evaluations. Site-level integrity teams are responsible for implementing the CML re-baseline activities and ensuring that all digital data uploads adhere to the naming conventions defined in this revision.

## Inspection Methodologies and Data Management

Phased Array Ultrasonic Testing (PAUT) is now the required method for corrosion mapping when high-resolution wall-loss data is necessary for integrity assessments. This transition from the methods described in QSCP-CIS-001 Rev 0 allows for more precise characterization of damage mechanisms identified through the application of API RP 571.

Digital data integrity is paramount; all inspection results must be recorded using the standardized CML-id rules to facilitate longitudinal data analysis. The use of PAUT for corrosion mapping provides the necessary data density to support advanced engineering evaluations and ensures consistency across the corporate fleet.

## Integrity Assessment and Fitness-For-Service

When inspection results indicate wall loss or cracking exceeding the original design margins, a formal assessment must be conducted. These evaluations shall follow the procedures outlined in API 579-1/ASME FFS-1 to determine the remaining life and continued serviceability of the piping components.

Dr. Helen Marchetti shall oversee the validation of FFS reports to ensure that the engineering logic aligns with corporate risk tolerances. All assessments must utilize the digital data collected under the new CML-id rules to ensure that the most accurate and recent wall thickness measurements are applied to the calculations.

## Referenced Standards and Compliance

The execution of piping inspections must strictly adhere to the requirements of API 570 for in-service systems. Damage mechanism identification shall be guided by API RP 571 to ensure that the appropriate NDE techniques, such as PAUT, are deployed against specific threats like localized corrosion or environmental cracking.

Certification bodies recognized under CAN/CGSB-48.9712 and EN ISO 9712 provide the necessary assurance of technician proficiency. All integrity activities, including the transition from the requirements of QSCP-CIS-001 Rev 0, must be documented within the corporate asset management software to maintain a continuous and auditable record of asset health.

| Sign-off | Name | Function | Date | Signature |
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| Prepared | Dr. Helen Marchetti | Corporate Asset Integrity Manager, P.Eng. | 2022-03-01 | _____ |